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EMPIRICAL RESEARCH

What Makes for a Vitalizing Day in Adolescence? Antecedents and Outcomes of Daily Need Crafting

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Abstract
Psychological need crafting denotes individuals' pro-active attempts to fulfill their psychological need for autonomy, relatedness, and competence. Although previous research has shown that need crafting contributes to adolescents' mental health, little is known about the factors involved in adolescents' daily efforts to engage in need crafting. This study considers psychological energy as both an important prerequisite for need crafting and as an outcome of daily need crafting. The sample consisted of 168 adolescents, aged 16–18 years ($M_{age} = 16.63$; 76.1% female). Morning and evening measurements were completed for 7 consecutive days. Adolescents' need crafting intentions in the morning were associated positively with vitality at the end of the day, an effect occurring through satisfaction of the basic psychological needs. Further, better sleep and subsequent morning vitality predicted more need crafting intentions in the morning. The findings indicate that both a psychological pathway (i.e., need crafting) and physical pathway (i.e., sleep) are critical to preserve adolescents' daily vitality.

Keywords Self-Determination Theory · Need crafting · Daily energy · Sleep · Adolescents

Introduction

According to Self-Determination Theory (SDT; Ryan and Deci, 2017), adolescents feel more energetic and alive (i.e., vitality) when they experience a sense of autonomy, relatedness, and competence (Frederick and Ryan, 2023). The pro-active search for activities, relational partners, and contexts that enable one to get these three psychological needs met is called need crafting (Laporte et al., 2021). Initial research suggests that need crafting is an important source of psychological need satisfaction for adolescents on top of the experienced parental support of their psychological needs (Laporte et al., 2021). Moreover, adolescents' need crafting has been shown to fluctuate from day-to-day, with corresponding day-to-day variations in positive and negative affect (Laporte et al., 2021). However, the exact role of need crafting in adolescents' daily functioning is still poorly understood. One unaddressed question is whether need crafting intentions early in the day play a directional role in adolescents' vitality later during the day. Further, not much is known about the reasons why on some days adolescents succeed in mobilizing their need crafting skills, whereas on other days they craft their needs much less. Herein, sleep and subsequent morning vitality could play a role in the degree to which adolescents formulate need crafting intentions in the morning. The overall aim of this study is to gain a deeper understanding of the antecedents and outcomes of adolescents' daily need crafting, thereby considering psychological energy both as an important effect and as a prerequisite for daily need crafting.

Basic Psychological Needs

SDT is a macro-theory on motivation, social development, and mental health. At the heart of this theory is the assumption that satisfaction of three basic psychological needs, that is, the needs for autonomy, relatedness, and competence, is a necessary prerequisite for well-being (Vansteenkiste et al., 2020). Autonomy satisfaction entails the experience of choice, psychological freedom, and authenticity. Autonomy frustration occurs when adolescents feel pushed in unwanted directions, experiencing pressure to feel, think, and act in a certain way. The need for relatedness denotes the experience of warmth, mutual care and a sense of belongingness to a group or community. When frustrated in this need, adolescents experience exclusion, loneliness and rejection. Competence refers to the

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experience of effectiveness and mastery, with adolescents being able to attain valued goals and to develop their skills and talents. When adolescents feel frustrated in this need, they experience a sense of failure and inadequacy.

Although the three basic psychological needs are assumed to be fundamentally important for adolescents' mental health, many adolescents struggle to get their needs met on a day-to-day basis (Soenens and Vansteenkiste, 2023). Adolescents typically want to take more responsibility for their own day-to-day activities (e.g., planning their schoolwork and meeting friends), yet they sometimes face demanding parents who want to monitor all of their whereabouts (e.g., social media use), entailing autonomy frustration. Further, although adolescents acquire new knowledge or skills at school and during leisure time activity, the development of competence often involves temporary failures they need to cope with. Finally, while adolescents' social network extends gradually, they also find out that the development of stable and intimate relationships requires time, patience, and trust. Said differently, adolescence is a period with both exciting new opportunities for need satisfaction, but pitfalls and threats to their basic needs.

Dozens of studies have demonstrated the crucial role of adolescents' basic needs for their psychological functioning, both at the between- and within-person level. Specifically, differences between adolescents in need satisfaction predict between-person differences in positive affect (Véronneau et al., 2005), resilience (Lera and Abualkibash, 2022) and life satisfaction (Eryilmaz, 2012), whereas differences in need frustration predict between-person differences in behavioral problems (Rodríguez-Meirinhos et al., 2020), depressive symptoms (Campbell et al., 2018), and burn-out symptoms (Warburton et al., 2020). At the within-person level, diary studies among adolescents indicate that daily fluctuations in need satisfaction correspond with daily fluctuations in self-esteem (Heppner et al., 2008) and well-being (Thomaes et al., 2017), and that day-to-day fluctuations in need frustration relate to daily variations in depressive symptoms (Vandenkerckhove et al., 2021), stress and subjective sleep quality (Campbell et al., 2021) and bulimic symptoms (Verstuyf et al., 2013).

Need Crafting

To get their basic needs fulfilled, adolescents benefit from being surrounded by need-supportive socializing agents, like teachers (Reeve and Cheon, 2021), parents (Mageau et al., 2022) or sport coaches (Morbée et al., 2024). Yet, as adolescence is a life period during which adolescents increasingly take control of their own lives, a crucial question is whether they may benefit from investing in their own need-based experiences (Soenens et al., 2019). In recent research, there has been increasing interest in the notion of need crafting, which is defined as the skill to proactively shape one's own psychological basic needs by searching for circumstances, activities and persons that have the potential to fulfill the psychological needs for autonomy, relatedness and competence (Laporte et al., 2021). Need crafting consists of an *awareness* component, that involves self-knowledge and clarity concerning the circumstances, activities, and individuals that may contribute to fulfilling one's needs. Need crafting also contains an *action* component, with individuals undertaking specific action to effectively get their needs fulfilled. To illustrate, adolescents who deliberately choose to play a challenging videogame (instead of scrolling mindlessly through social media feeds) engage in autonomy crafting. Adolescents who go out for running with a sibling to spend some time together could do so to experience greater connection and, hence, involves relatedness need crafting. Finally, adolescents who make up a realistic study plan to perform well on a series of exams are investing in competence crafting.

The examples above make clear that crafting of one need often co-occurs with the crafting of other needs. Consistent with this observation, research with a recently developed scale for need crafting (Laporte et al., 2021) showed that the three need crafting subscales were substantially correlated, justifying the creation of a composite score. Testifying to its validity, the overall score for adolescents' need crafting was associated positively with related constructs denoting a self-reflective and agentic orientation, including mindfulness, pro-active personality, and agentic engagement. In line with the hypothesis that need crafting contributes to the quality of adolescents' need-based experiences and mental health, need crafting was found to relate positively to need satisfaction and well-being, a finding observed both at the between-person level and at the within-person level (Laporte et al., 2021). In addition, a diary study among adolescents showed that need crafting varied considerably from day to day (Laporte et al., 2021). On evenings that adolescents reported that they had crafted their needs more, they also reported corresponding higher need satisfaction and positive affect and lower need frustration and negative affect.

One important limitation of previous studies is that need crafting and adolescents' outcomes were always measured concurrently (e.g., in the evening). This concurrent assessment leaves the question unaddressed whether need crafting *prospectively* feeds into improved need satisfaction. An alternative possibility is that adolescents' need crafting is merely a by-product of their encountered need satisfaction that day. Specifically, responses to the scale measuring need-based experiences may affect the responses on the need crafting scale (e.g., "My needs are satisfied, so I probably did something myself to experience more need satisfaction"). Higher scores on the need crafting scale would then represent a post-hoc attribution of one's scores on the scale for need-based experiences. To determine with greater certainty that need crafting actually contributes to later need-based experiences, the assessment of need crafting should be decoupled in time from the assessment of need-based experiences. By measuring adolescents' need crafting intentions in the morning and their daily need-based experiences and vitality in the evening (thereby even controlling for adolescents' need-based experiences the day before), this study sheds light on the prospective association between need crafting, need-based experiences, and vitality. Indirect evidence for this assumed direction of effects comes from research showing that a brief online intervention in need crafting results in improved need satisfaction and mental health amongst adults (Laporte et al., 2024) and students (van den Bogaard et al., 2024a).

The Role of Sleep and Energy in Need Crafting

Need crafting is assumed to energize people's behavior, leading them to take pro-active action (Laporte et al., 2021). At this time, this skill requires mental efforts and engagement in activities and, as such needs to be energized itself. Therefore, an important next question is which factors can explain why adolescents mobilize their need crafting skills more on some days than on others? In addition to time constraints and contextual demands that may limit opportunities for need crafting, adolescents' available energy, as indexed by the presence of vigor and the absence of feelings of depletion (Frederick and Ryan, 2023), may play a role. Whereas prior SDT-based research has primarily treated energy-related variables as outcomes of need satisfaction (e.g., Vansteenkiste et al. 2006), energy may also serve as a predictor of need crafting and subsequent need satisfaction. Vitality refers to the (physical and mental) energy that is available to the self (Ryan and Frederick, 1997), with such energy being essential for self-reflection and engagement in need-conducive activities (Nix et al., 1999).

In turn, adolescents' energy is likely affected by their sleep quality. Sleep indeed serves as a well-known supplier of morning vitality through the restoration of cognitive resources (Eugene and Masiak, 2015). An examination of adolescents' sleep in relation to energy and need crafting is important because adolescence is a developmental period of risk for sleeping problems due to biological changes (Carskadon, 2008), academic (Wolfson and Carskadon, 1998) and extracurricular demands (Roberts et al., 2011), and sleep-interfering habits such as excessive media use (Alonzo et al., 2021). Both adolescents' sleep quantity (i.e., objective or self-reported hours of effective sleep) and their sleep quality (i.e., felt restorative nature of sleep) have been extensively studied as predictors of their mental health and functioning, predicting mood deficits (Short et al., 2020), substance use (Wahlstrom et al., 2017), fatigue and difficulties to get up in the morning (Gau and Soong, 1995), and disrupted cognitive functioning (de Bruin et al., 2017). Sleep quality thereby often yields stronger associations with health-related outcomes than sleep quantity. A recent meta-analysis (O'Callaghan et al., 2021) showed that subjective sleep quality had a stronger negative association with depressive symptoms compared to sleep duration in adolescence and a longitudinal study indicated that higher sleep quality, but not sleep quantity, predicted enhanced well-being and self-esteem of adolescents one year later (Vazsonyi et al., 2021).

Within SDT, a growing number of studies have shed light on the interplay between sleep, energy and need-based experiences (see Campbell and Vansteenkiste 2023 for an overview). A diary study among adult patients with chronic fatigue syndrome showed that day-to-day variations in sleep quality, as reported in the morning, predicted day-to-day variations in need satisfaction and need frustration in the evening, an effect that was partly mediated by enhanced fatigue and reduced vitality in the morning (Campbell et al., 2018). A similar dynamic process was observed in a 7-day diary study among adolescents (Campbell et al., 2021). Self-reported (but not observed) sleep quantity and subjective sleep quality in the morning related positively to need satisfaction and negatively to need frustration in the evening, with morning fatigue serving as the mediating mechanism.

Herein, it is argued that need crafting might serve as a missing link in associations between sleep/energy and need-based experiences, helping to explain why poor sleep and morning fatigue predict less satisfaction and more need frustration at night. Two earlier studies provide indirect evidence for this reasoning. First, in an experimental study, which required participants to limit their sleep to five hours during three consecutive days, sleep deprivation was found to increase morning fatigue the next day. This enhanced fatigue then impaired participants' mindfulness, a resource needed to preserve need satisfaction (Campbell et al., 2018). Because mindfulness is positively related to need crafting (Laporte et al., 2021), it can be expected that need-crafting intentions may also be forestalled when fatigue is elevated in the morning after a poor or short night of sleep. Second, a 7-day diary study (Schmitt et al., 2017) among employees showed that sleep quality (but not quantity) was related to more daily vitality and subsequent pro-active behavior at work, which can be seen as a manifestation of need-crafting behavior at work. Together, these studies provide initial and indirect evidence for sleep (i.e., quantity and quality) and energy (i.e., vitality and depletion) as crucial conditions for the formulation of need crafting intentions.

Current Study

Although evidence suggests that need crafting relates to better mental health among adolescents, the precise role of need crafting in adolescents' daily functioning is not well understood. The objectives of the present diary study are to gain a better understanding of, first, the benefits of adolescents' need crafting during the day and, second, the role sleep and morning vitality in need-crafting intentions. First, on days when participants formulated higher need crafting intentions in the morning, they would report higher daily vitality and lower depletion in the evening because they experienced more need satisfaction and less need frustration that day (Hypothesis 1). Second, participants are expected to formulate stronger need-crafting intentions in the morning on days when they reported better sleep quality and longer sleep duration, due to the energizing effect (higher vitality and lower depletion) of sleep (Hypothesis 2). Hypotheses are situated at the within-person level (i.e., the level of daily variations).

Method

Participants and Procedure

A total of 181 Dutch-speaking Belgian adolescents participated initially in this study, of whom 13 participants were excluded from the dataset because they did not fill out any of the diary assessments. From the remaining participants ($N = 168$; $M_{age} = 16.63$, $SD = 0.60$, range = 16–18; 76.1% female; 57.9% academic track, 30.2% a technical track and 10.1% a vocational track), 57.14% filled out all 14 measurements (i.e., 7 morning and 7 evening measurements). In total 13.72% of all data was missing. Missingness was greater during evening measurements (ranging from 9.55% to 25.06%) compared to morning measurements (ranging from 4.03% to 17.59%). More missingness was observed to the end of the week, reaching its peak on Sunday for both morning and evening measurements. See Table 3 in the supplementary material for more details. Little's (1988) missing completely at random (MCAR) test on the variables of interest yielded a normed chi-square of 1.88, indicating that missing data could be reliably estimated (Bollen, 1989). Three impossible values on the sleep quantity measure (i.e., a negative amount of sleeping hours) were identified and replaced with missing values.

An a priori power analysis was not performed because no previous studies examined daily associations between adolescents' need crafting, vitality, and sleep using a similar design. Yet, previous diary studies with related constructs (e.g., psychological need satisfaction) suggested that the present study was sufficiently powered. A sample size of 60 students and 14 measurement points (Sheldon et al., 1996) and a sample size of 211 adolescents and 2 assessments across 8 days (Campbell et al., 2021) were sufficient to detect significant effects of need-based experiences on, respectively, daily well-being and daily sleep patterns. Also, a Monte Carlo simulation on two-level models with effects and input parameters typical for psychological research and a power of ≥0.80, showed that medium effects can be detected with measurements points ranging from 2 to 30 and sample sizes ranging from 30 to 200 (Arend and Schäfer, 2019). The present study, involving 168 participants and 14 measurement moments, falls within this range.

Data were collected by 181 undergraduate students, each of whom recruited one participant (between 16 and 18 years old) to whom they had no close relationship (e.g., family members, pupils from their youth association). Recruitment and follow-up of the participants were part of a course on developmental psychology, during which students learned about the different steps involved in research. Students made a choice between three themes (i.e., need crafting, gaming, and parental voice). Before recruitment took place, students were introduced to the theme, received a manual with step-by-step explanations of research procedures, including standardized information to convey to participants. Additionally, students were trained in a 90-min session with the first author. This training session prepared students for the home visit and daily follow-up, including various scenarios they could potentially encounter. During this session, students also received information about the ethical aspects of recruitment and interactions with participants. All students signed a confidentiality clause before the home visit, declaring they would not share any information obtained from the participant with others. After this session, the first author remained available for further questions through e-mail. If students were not able to find a participant for legitimate reasons, they were offered an alternative task to pass the course.

During the home visit students explained the purpose and set-up of the study and asked for the preferred way to be followed on a daily basis (i.e., via WhatsApp, e-mail, etc.). All participants and their caregivers were informed about their rights and data protection guidelines during the home visit (i.e., informed consent). The diary study started on Monday after the students' house visit and took place during a regular school week. Links to the questionnaires were distributed via e-mail. Participants were asked to fill out a short online questionnaire directly after waking up and before going to bed for seven successive days, either on their mobile phone or laptop. All morning assessments and all evening assessments had the same content, presented in the same order. Approval for this study was obtained from the faculty's Ethical Committee (2021/113).

Table 1 Descriptive statistics, ICC, McDonald's omega, and correlations within- and between- person level variables

	M	SD	ICC	ω-within (CI)	ω-between (CI)	1.	2.	3.	4.	5.	6.	7.	8.	9.
Morning														
1. Sleep quality	70.79	17.89	0.38	–	–		0.34***	0.49***	–0.42***	0.43***	0.34***	–0.27***	0.45***	–0.38***
2. Sleep quantity	474.47	99.19	0.19	–	–	0.47***		0.20*	–0.22***	0.06	0.14	–0.03	0.11	–0.16*
3. Vitality	2.73	1.04	0.37	0.87 (0.86–0.89)	0.98 (0.97–0.99)	0.39***	0.24***		–0.69***	0.49***	0.46***	–0.39***	0.70***	–0.57***
4. Depletion	2.56	1.04	0.37	0.80 (0.77–0.82)	0.97 (0.96–0.99)	–0.42***	–0.31***	–0.72***		–0.33***	–0.44***	0.51***	–0.56***	0.82***
5. Need crafting intentions	3.81	0.7	0.43	0.74 (0.72–0.77)	0.96 (0.94–0.97)	0.08*	–0.05	0.26***	–0.19***		0.74***	–0.46***	0.52***	–0.43***
Evening														
6. Need satisfaction	3.67	0.69	0.44	0.75 (0.72–0.78)	0.94 (0.92–0.96)	0.12***	0.03	0.27***	–0.22***	0.45***		–0.71***	0.62***	–0.58***
7. Need frustration	2.34	0.75	0.49	0.69 (0.65–0.72)	0.88 (0.84–0.92)	–0.08*	–0.04	–0.25***	0.19***	–0.27***	–0.57***		–0.45***	0.60***
8. Vitality	3.2	0.97	0.40	0.84 (0.82–0.86)	0.98 (0.96 –0.99)	0.25***	0.15***	0.44***	–0.38***	0.34***	0.52***	–0.40***		–0.71***
9. Depletion	2.35	0.98	0.39	0.74 (0.71–0.77)	0.96 (0.94–0.98)	–0.28***	–0.21***	–0.40***	0.45***	–0.25***	–0.41***	0.35***	–0.69***	

Under diagonal refers to within-subject correlations, above diagonal refers to between-subject correlations

*$p < 0.05$, ***$p < 0.001$

Measurements

Socio-demographic Variables

Participants indicated their gender (i.e., girl, boy, or different), age (i.e.,16, 17, 18, or specified their age when none of these categories were applicable), and the educational track they were following (i.e., vocational, technical or academic).

Morning Measures

All morning and evening constructs showed sufficient within and between-person scale reliabilities, as assed by McDonald's omega (ω) (Hayes and Coutts, 2020), see Table 1.

Sleep

Sleep quality was assessed using a visual analogue scale (VAS) from the Pittsburgh Sleep Diary (Monk et al., 1994). Participants rated their previous night's subjective sleep quality (i.e., "How was your sleep quality last night?") on a scale from 0 (*extremely poor*) to 100 (*extremely good*). Sleep quantity was calculated using four open-ended questions from the Pittsburgh Sleep Diary (Monk et al., 1994), which assessed evening bed-time, morning wake-time, number of minutes it took to fall asleep (sleep latency), and the number of minutes spent awake during the night after initial sleep onset (wake after sleep onset). Total time in bed was calculated on the basis of the bed and wake times. Similar to other studies (Campbell et al., 2021), daily sleep duration was calculated by subtracting sleep latency and wake after sleep onset from the total time spent in bed.

Need crafting intentions

Need crafting intentions were assessed through a set of six items (i.e., 2 items per need) from the Dutch version of the Need Crafting Scale (NCS: Laporte et al., 2021). Items were slightly adapted to make them amenable for a diary format, thereby focusing on participants' intended actions. An example for competence crafting is: *"I intend to do something today that I am good at"*. Items were rated on a 5-point Likert scale ranging from 1 (*completely not true*) to 5 (*completely true*).

Vitality and depletion

Morning vitality (e.g., "At this moment, I feel energetic") was assessed using three items of the Subjective Vitality Measurement (SVM; Ryan and Frederick, 1997), scored on a 5-point Likert scale ranging from 1 (*not at all*) to 5 (*very* *much)*. Morning vitality showed an average reliability of $\alpha = 0.90$, range across the measurements (range = 0.88–0.92). Morning depletion (e.g., "I feel exhausted at this moment") was assessed using three items of the Lassitude subscale from the Inventory of Depression and Anxiety Symptoms (IDAS; Watson et al., 2007). Items were scored on a 5-point Likert scale ranging from 1 (*not at all*) to 5 (*very much)*.

Evening Measures

Vitality and depletion

The same set of items was used to assess evening vitality and depletion in the evening, with the stem being slightly adjusted (e.g., "Today I felt energetic" for vitality and "Today I felt exhausted" for depletion).

Satisfaction and frustration of the basic needs

Need-based experiences were measured with the 12-item diary version of the Basic Psychological Need Satisfaction Need Frustration-scale (BPNSNF; Chen et al., 2015). Psychological need frustration (e.g. "Today, most things I did felt like "I had to", for autonomy frustration) and need satisfaction (e.g. "Today, I felt confident I did things well", for competence satisfaction) were both captured by two items per need (six items per scale in total). Items were rated on a scale ranging from 1 (*completely not true*) to 5 (*completely true*).

Plan of Analysis

As this study consisted of repeated measurements on seven consecutive days (i.e., within-level) nested within 168 adolescents (i.e., between-level), multilevel models (including calculation of all indirect and total effects) were performed with Lavaan (Rosseel, 2012) in R 4.3.1, using MLR as estimator. Model fit was evaluated using a combination of the Root-Mean-Square Error of Approximation (RMSEA), the Standardized-Root-Square Residual (SRMR) and the Comparative Fit Index (CFI). Values of a RMSEA value below 0.08, a SRMR value below 0.06 and a CFI value of 0.90 or more are indicative of a good model fit (Hu and Bentler, 1999). Prior to investigating the main hypotheses, intra-class correlations (ICC) were calculated to check whether the study variables showed sufficient daily variability and to examine whether multilevel modeling was needed to decompose the total variance in all study variables into variance due to differences at the between-person level and at the day-to-day or within-person level.

Two-level path models were estimated to test the key hypotheses at the within-person level, while controlling for the variance in the study variables at the between-person level. The first model tested Hypothesis 1, according to which daily need crafting intentions in the morning relate to energy (i.e., vitality and depletion) in the evening, mediated by need satisfaction and need frustration respectively. To estimate changes in reported vitality and depletion during the day, the evening measures of these variables were controlled for vitality and depletion in the morning, respectively. To provide a conservative test of the role of morning need-crafting intentions in evening need-based experiences, an additional model was tested that controlled for need-based experiences the day before (which included one day of measurements less than the basic model).

A second two-level path model was tested to examine whether daily within-person variations in the morning reports of need crafting intentions are predicted by quality and quantity of sleep, through experienced vitality and depletion (Hypothesis 2). Sleep quantity and sleep quality were standardized for these analyses due to their extreme mean and standard deviations compared to the other variables. In a second step the model was controlled for need crafting intentions the day before, to again obtain a more conservative test of the assumed daily spiral (and with this model again including one day of measurements less than the basic model).

Results

Preliminary Analyses

The intra class correlation (ICC) indicated that all constructs showed enough variation at the two different levels of analysis to apply multilevel modeling (above 0.05, Preacher et al., 2010). The ICC indicates the percentage of variance situated at the between-person level. Need crafting intentions had an ICC of 0.43, indicating that 43% of the variance is situated at the between person level, with 57% of the variance thus being situated at the within person level. It should be noted that the variance at the within-person level also includes measurement error. The ICCs for the other constructs were as follows: 0.38 for sleep quality, 0.19 for sleep quality, 0.37 and 0.40 for, respectively, morning and evening vitality, 0.37 and 0.39 for, respectively, morning and evening depletion, 0.45 for need satisfaction and 0.49 for need frustration.

Table 1 presents McDonald's omega, means, standard deviations and correlations between the study variables at the within- and between-person level. Correlations were found to be generally consistent with theoretical assumptions. Daily sleep quality and quantity related positively to morning and evening vitality and negatively to morning and evening depletion. The correlations with the need-relevant

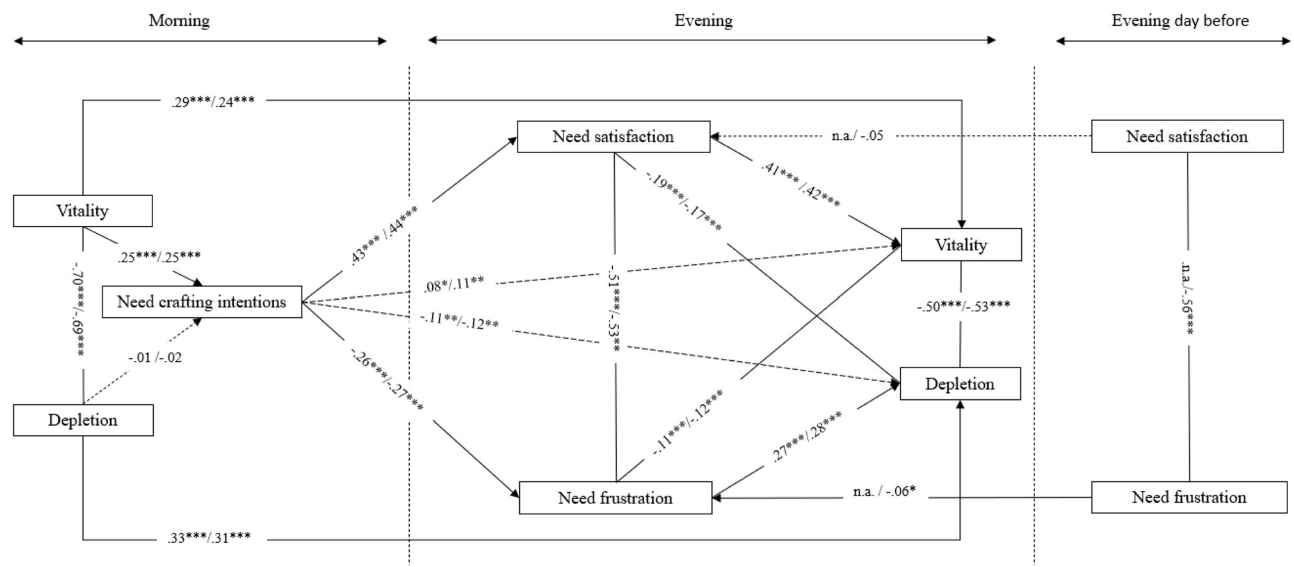

Fig. 1 Within-person Level Results of a Multi-level Mediation Model (Hypothesis 1). *Note*. Direct effects after adding Need crafting intentions the day before are reported after the slash "/", original direct effects are reported before the slash "/". Coefficients shown are standardized path coefficients. CFI = 0.97/1.00, RMSEA = 0.08/0.00, SRMR= 0.06/0.06. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

measures were less strong, with only sleep quality (and not sleep quantity) being positively associated with need crafting intentions and need satisfaction while being negatively correlated with need frustration.

In a next step, a multilevel model was set up to examine the association between type of day (weekday versus weekend day) as a within-person predictor and with various socio-demographic background variables (i.e., age, gender, and education) as between-person predictors of the study variables. At the within-person level, a significant effect was found for weekday (versus weekend -day) on need crafting intentions ($b = 0.13$, $SE = 0.049$, $p < 0.05$), vitality (morning/evening) ($b = 0.28/0.24$, $SE = 0.082/0.053$, $p < 0.01/0.000$), depletion (morning/evening) ($b = –0.32/–0.18$, $SE = 0.077/0.059$, $p < 0.000/0.01$), need crafting ($b = 0.21$, $SE = 0.037$, $p < 0.000$), need satisfaction ($b = 0.18$, $SE = 0.043$, $p < 0.000$), and need frustration ($b = –0.15$, $SE = 0.047$, $p < 0.01$). During the weekend, participants report more need crafting intentions, morning and evening vitality and need satisfaction and less morning and evening depletion and need frustration. At the between-person level, no significant effects were found of adolescents' age. A gender effect was found for both morning and evening vitality ($b = 0.56/0.45$, $SE = 0.131/0.119$, $p < 0.000$), for both morning and evening depletion ($b = –0.39/–0.27$, $SE = 0.128/0.117$, $p < 0.01/ < 0.05$) and for need frustration ($b = –0.26$ $SE = 0.104$, $p < 0.05$). Girls reported less vitality, more depletion and more need frustration than boys. Also, a significant effect was found for type of education on morning vitality ($b = 0.15$, $SE = 0.075$, $p < 0.05$), with adolescents following an academic track reporting more vitality than adolescents following a vocational track. Subsequent analyses are therefore controlled for adolescents' gender and education at the between-person level and for type of day (i.e., weekday or weekend day) at the within-person level.

Primary Analyses

The first model, with need crafting intentions in the morning predicting need-based experiences and felt energy in the evening while controlling for morning measures of energy can be found in Fig. 1. Fit indices show good fit (CFI = 0.97, RMSEA = 0.08, SRMR = 0.06). As shown in Table 2, the paths from need-crafting intentions to need-based experiences (i.e., need satisfaction and to need frustration), and the paths from need-based experiences to energy (i.e., vitality and depletion) were significant. Even with need satisfaction and need frustration included as intervening variables, the direct paths from need crafting intentions to vitality and depletion remained significant. All indirect effects were significant, indicating that need crafting intentions related positively to evening vitality through evening need satisfaction, and related negatively to evening depletion through evening need frustration.

Two additional findings should be highlighted. First, this model controlled for morning vitality and depletion, suggesting that need satisfaction predicts a within-person increase in vitality and need frustration predicts a within-person increase in depletion. Second, when controlled for need-based experiences of the day before in a truncated model (CFI = 1.00, RMSEA = 0.00, SRMR = 0.06), all paths remained significant, contributing to the robustness of the daily sequence presented. The coefficients of the latter, more conservative, model are shown in Fig. 1 after the slash.

The second hypothesized model included paths from sleep and morning energy to morning need crafting intentions (Fig. 2). Fit indices show good fit (CFI = 1.00, RMSEA = 0.04, SRMR = 0.00). An overview of the path coefficients can be found in Table 2. All paths from sleep (i.e., sleep quantity and sleep quality) to energy (i.e., vitality and depletion) were significant. As for the prediction of need crafting intention, vitality yielded a significant positive association, whereas a non-significant path was found from depletion to need crafting intentions. Need crafting intentions thus were mainly predicted by sleep quality, through vitality, as indicated by the only significant indirect effect of this model (see Table 2). The indirect path from sleep quantity to need crafting intentions through vitality, did not reach significance. As for the direct effect from sleep to need crafting intentions, only sleep quantity exerted a direct effect on need crafting intentions. Unexpectedly this direct path was negative, indicating that more sleep quantity predicted less need crafting intentions. All other directions of the coefficients were in line with the hypothesis (e.g., sleep quality related positively to vitality, that in turn related positively to need crafting intentions). After controlling for need crafting intentions of the day before in a truncated model (CFI = 1.00, RMSEA = 0.00, SRMR = 0.02), most associations remained significant. Only one effect was no longer significant, that is, the path from sleep quantity to morning vitality.

Table 2 Standardized path coefficients of mediation models (Hypothesis 1 & 2)

Paths	Indirect effects β
Model 1a/Model 1b	
Need crafting intentions-Need satisfaction-Vitality	0.18***/0.18***
Need crafting intentions-Need frustration-Depletion	–0.07***/–0.08***
Model 2a/Model 2b	
Sleep quality-Vitality-Need crafting intentions	0.08**/0.09**
Sleep quality-Depletion-Need crafting intentions	0.01/0.02
Sleep quantity-Vitality-Need crafting intentions	0.02/0.02
Sleep quantity-Depletion-Need crafting intentions	0.01/0.01

Coefficients shown are standardized path coefficients (β)

$^{**}p < 0.01$, $^{***}p < 0.001$

Sensitivity Analysis

Sensitivity analyses were performed by testing both models per basic need (i.e., autonomy, relatedness, competence) instead of using the composite scores for need satisfaction. Overall, these additional models indicated that findings generalize across the three needs, thereby testifying to the robustness of the tested pathways. Some small differences were noted between the three needs. Specifically, in Model 1a, the negative association between need crafting intentions and need frustration was significant only for autonomy (and not for the other two needs). In Model 2a, there was a direct negative association between sleep quantity and need crafting intentions (beyond morning vitality and depletion) only for the need for relatedness (and not for the other two needs). Finally, competence need crafting intentions were not related to vitality in either models. These exceptions

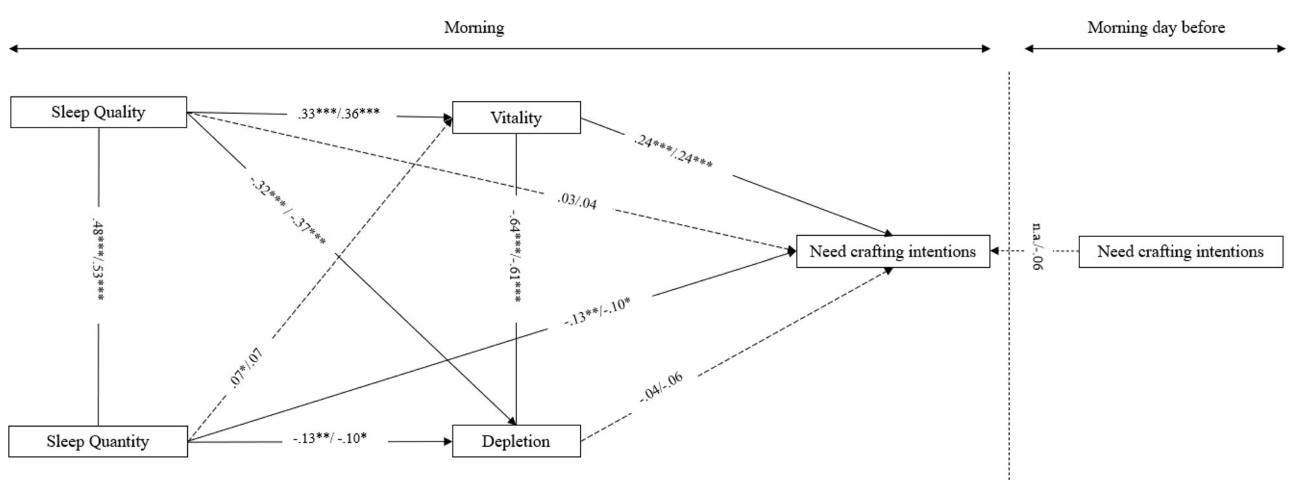

Fig. 2 Within-person Level Results of Multi-level Mediation Model (Hypothesis 2). *Note*. Direct effects after adding Need crafting intentions the day before are reported after the slash "/", original direct effects are reported before the slash "/". Coefficients shown are standardized path coefficients. CFI = 1.00/1.00; RMSEA = 0.04/0.00; SRMR = 0.00/0.02. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

should be interpreted with caution because the separate needs were assessed with just two items. See Figure 3–8 in supplementary material for more details.

Discussion

Research based on SDT (Ryan and Deci, 2017) has shown that satisfaction of the basic psychological needs for autonomy, competence, and relatedness is crucial for adolescents to thrive (Soenens and Vansteenkiste, 2023) and experience daily well-being (Thomaes et al., 2017). To understand where individual and within-person differences in need satisfaction come from, most previous studies have focused on the need supportive (or thwarting) role of adolescents' social context. Recent research, however, has shown that adolescents can also play a role themselves in fulfilling their basic psychological needs through the capacity for need crafting. The present 7-day diary study among adolescents, aimed to contribute to this emerging research field in two ways. First, this study examined whether need-crafting intentions in the morning predict need-based experiences and energy later during the day, thereby ensuring temporal precedence in the assessment of need crafting to its presumed outcomes. Second, the current study aimed to better understand the conditions accounting for daily fluctuations in need crafting and associated well-being indicators, thereby focusing on the role of adolescents' sleep and energy.

Key Findings and Implications

Multilevel analyses confirmed the hypothesis that need crafting intentions in the morning are positively related to vitality and negatively related to depletion in the evening, via respectively need satisfaction and need frustration. These associations remained significant after controlling for need satisfaction and need frustration the day before, suggesting that need crafting intentions even result in heightened need satisfaction compared to the previous day. These findings increase confidence in the assumed directional role of need crafting in adolescents' need-based experiences and vitality.

The next research question was about the role of sleep and morning energy in adolescents' need-crafting intentions. Results showed that need-crafting intentions in the morning were predicted by sleep quality via morning vitality, whereas depletion did not play a role in adolescents' need-crafting intentions. These findings suggest that the absence of depletion is not enough to ignite a positive spiral. Instead, morning vitality seems to be a prerequisite to formulate need-based intentions for the day ahead. Unexpectedly, sleep quantity was related slightly negatively to need crafting intentions. Adolescents who sleep longer, may more easily run out of time in the morning, resulting in less reflection on their need-crafting intentions. It is also possible that sleeping longer is indicative of depletion and ill-being. Indeed, research has shown a U-shaped relation between sleep duration and mental health problems (Cappuccio et al., 2010). The association between sleep quality and need-crafting intentions through vitality, remained significant after controlling for need-crafting intentions the day before, suggesting that a subjective good night of sleep results in increased formulation of need-crafting intentions compared to the day before.

The finding that sleep quality yields stronger associations with need crafting than sleep quantity is consistent with previous research showing that sleep quality is related more strongly to mental health than average sleep quantity (Pilcher et al., 1997), and that daily sleep quality, but not sleep quantity, is positively related to more daily need satisfaction and less daily need frustration through daily energy (Campbell et al., 2018).

Overall, the findings suggest that a positive spiral can unfold in adolescents' daily functioning. After a good night of sleep, adolescents have more mental energy to engage in need crafting, with need crafting in turn leading to more need satisfaction and vitality by the end of the day. Because previous research (Campbell et al., 2021) has shown that adolescents sleep better after a need-satisfying day, adolescents are likely to again engage in need-crafting the next day. Interpreted the other way around, poor sleep and a lack of vitality in the morning seem to set the stage for a downward spiral, with adolescents engaging less in need crafting and feeling more depleted by the end of the day.

Given the crucial role of need-based experiences for adolescents' energy and mental health, this theme deserves being addressed at a larger scale in schools. Schools can invest in the monitoring of students' basic needs through regular school-wide surveys and subsequently make the topic of need crafting central to one-to-one teacher-student conversations. Student can also be trained in the topic of need crafting. One particularly helpful tool in this regard could be LifeCraft (Laporte et al., 2024), an online intervention program aimed at fostering need crafting. The effectiveness of this program has been demonstrated to promote students' psychological need satisfaction and mental health during exams by training students during (van den Bogaard et al., 2024a) and one month prior to their exams (van den Bogaard et al., 2024b). This program can help inform student counsellors and teachers about the theme of need crafting and can be offered directly to students to enhance their need-based experiences.

As current findings highlight the important role of sleep to formulate need crafting intentions, it may be important to complement such need-crafting interventions with interventions that target adequate sleep. Research has shown that adolescents' sleep can be improved through cognitive-behavioral interventions, with such interventions having lasting effectiveness in enhancing subjective sleep quality and energy levels (Blake et al. 2017, 2019 for an overview). Such interventions typically include psycho-education, goal-setting, and exercises to better cope with sleep worries. They can be delivered in small training groups (e.g., Wolfson et al. 2015) or via online applications (e.g., Carmona et al. 2021). An important goal for future research is to examine whether the combination of an intervention focusing on sleep quality and an intervention focusing on need crafting is more effective to improve adolescents' mental health than interventions focusing on only one of both target variables. By combining both types interventions, adolescents are not only taught the skill to engage in need crafting but are also provided with sufficient energy to implement this skill, possibly resulting in the positive spiral observed in the current study.

Since adolescents do not yet live independently, the current findings also imply indirectly that caregivers can play a role in adolescents' ability to generate energy throughout the day. The chain of events examined in this study suggests that caregivers can intervene through two different routes. First, they can encourage healthy sleep habits by setting rules for bedtime and by communicating about these bedtime rules in autonomy-supportive ways (Peltz et al., 2020). Second, caregivers could try to facilitate adolescents' need-crafting intentions, for example by having brief conversations in the morning (e.g., during breakfast) about the day ahead that may result in setting need-crafting intentions. It would be interesting for future research to examine whether caregivers can buffer the effects of poor sleep by facilitating adolescents' need-crafting intentions in the morning and need-crafting throughout the day.

Limitations and Directions for Future Research

Although this study had a number of methodological strengths (including the diary design and the inclusion of morning and evening measures), it also has a couple of limitations. First, the fact that undergraduate psychology students recruited the participating adolescents is reflected in the socio-demographic characteristics of the participants, with a majority following an academic track and being female. Future research is needed to replicate these findings in more heterogeneous and representative samples.

The generalizability of the results could also be examined across more diverse time periods. Whereas the current study was conducted during a regular school week, future research could examine whether similar dynamics operate during other periods such as exam periods or holidays. This is important because adolescents' mobilization of need-crafting skills probably depends on contextual demands and affordances, with holiday periods for instance providing more opportunities for need crafting than exam periods (van den Bogaard et al., 2024b). Research already showed that college students' need-based experiences are restored after exams (Campbell et al., 2018), an effect that could be due partly to increased opportunities for need crafting after the exams. Identifying factors that contribute to the retention of need-crafting skills throughout different periods that vary in terms of demandingness, would be of value to strengthen prevention and intervention programs.

A second limitation is the use of self-reported measures only. Although the measures of the predictors and outcomes were separated in time, associations between the variables may still have been inflated due to the self-reported nature of all measures. As sleep (i.e., quantity and quality) was measured with self-reports only, future research would do well to include objective sleep measurements, which have been found to predict need-based experiences (Campbell et al., 2021). Related to the limitation of self-reported study variables, no behavioral co-variates were included either. A relevant control variable to consider in future research is screen time. Perhaps excessive screen time hinders accomplishment of need-crafting intentions and reduces sleep quality (Nakshine et al., 2022), thereby explaining part of the association between adolescents' need-crafting intentions and energy.

A third limitation is the somewhat generic measure of need crafting intentions, which only tapped into the degree to which adolescents had intentions in the morning. This measure provides no information about exactly which activities the adolescents wanted to engage in during the day. To gain a better insight in the nature of adolescents' daily need-crafting activities, future diary studies could include more open-ended questions about participants' self-formulated need-crafting intentions in the morning. Adolescents' need-crafting intentions could then also be rated on criteria such as specificity, feasibility, and degree of fit with each of the three psychological needs, with such criteria perhaps reflecting the quality of their need-crafting efforts.

Lastly, although the inclusion of a morning and evening assessment was a strength compared to previous research, the use of only two assessments per day limits understanding of how dynamics involved in need crafting unfold during the day. The present study included no information about whether and how adolescents adjust their need-crafting intentions to unexpected setbacks (e.g., a last-minute cancellation from a friend) and opportunities (e.g., an invitation to join a group of peers at the mall). To examine these within-day dynamics to a fuller extent, Experience Sampling Methodology with multiple assessments per day could be useful (Myin-Germeys et al., 2009, 2018).

Conclusion

Not much is known about why on some days adolescents succeed in mobilizing the skill to pro-actively search for and engage in activities that satisfy the need for autonomy, competence and relatedness, while on other days it seems more difficult to engage in need crafting. Findings of the current 7-day diary study confirm that adolescents who craft their own needs generate energy throughout the day via basic psychological need satisfaction. However, energy also turned out to be a prerequisite for adolescents to initiate daily need crafting, with sleep quality of the night before and subsequent morning energy predicting more need-crafting intentions. To improve adolescents' ability to take control of their daily energy, findings indicate the importance of a psychological pathway (i.e., need crafting) and a physical pathway (i.e., sleep).

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Authors' contributions D.B. conceived of the study, participated in its design and coordination, performed the statistical analysis and interpretation of the data and drafted the manuscript; B.S. conceived of the study, participated in its design and interpretation of the data and helped up to draft the manuscript; K.B. participated in interpretation of the data and helped up to draft the manuscript; M.V. conceived of the study, participated in its design and interpretation of the data and helped up to draft the manuscript. All authors read and approved the final manuscript.

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Data sharing and declaration The datasets generated and/or analysed during the current study are available in the OSF repository, https://osf.io/5ynxc/.

Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

Ethical approval The research was conducted according to the ethical rules presented in the General Ethical Protocol of the Faculty of Psychology and Educational Sciences of Ghent University.

Informed consent All participants provided informed consent prior to participating in the study.

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